

328637

Course Name: Operating System

Course Outcome

- CO1- Understand the classification of operating system environment.
 CO2- Understand the basic of process management.
 CO3- Apply the concept of CPU process scheduling for the given scenarios.
 CO4- Illustrate the process synchronization and concurrency process in operating system.
 CO5- Analyze the occurrence of deadlock in operating system.
 CO6- Describe and analyze the memory management and its allocation policies.
 CO7- Understand the concepts of disk scheduling

Printed Pages: 04

University Roll No.

End Term Examination, Odd Semester 2025-26
BTech (CSE/AIML/EC(CS)/ECSE), II Year, III Semester
Operating Systems (BCSC0004)

Time: 3 Hours

Maximum Marks: 45

Section – A

Attempt All Questions

3 X 6 = 18 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Why can SSTF disk scheduling lead to starvation? Consider a disk queue with requests for I/O to blocks on cylinders 55, 58, 60, 70, 18, 90, 150, 160, 184. The cylinders are numbered from 0 to 199. The head is initially at cylinder number 80. Find total head movement using SSTF.	1+2	7	A	P
2	A System has three processes P1, P2, P3 and three resources: R1, R2, R3, each with one instance. The Current state is as follows: • P1 holds R1 and requests R2. • P2 holds R2 and requests R3. • P3 holds R3 and requests R1. • Draw the Resource Allocation Graph (RAG) • Convert it to the Wait-for Graph (WFG) • Is there deadlock?	1+1+1	5	U	C
3	A system has free memory blocks of 130 KB, 80 KB, 200 KB, 260 KB, and 140 KB. Five processes (in order) request memory in sizes 120 KB, 90 KB, 50 KB, 220 KB, and 100 KB. Allocate memory using First-Fit.	1+1+1	6	A	P

	Best-Fit, and Worst-Fit strategies.				
4	Explain Contiguous, Linked, and Indexed file allocation strategies. Discuss advantages and disadvantages of each.	1+1+1	7	R	F
5	Consider a machine with 64 KB Logical address space and 32 KB Physical address space. if the Page size is 4 KB then Calculate the following: a. Number of pages in the logical address space b. Number of frames in the physical memory c. Number of bits required for page number and offset.	1+1+1	6	A	P
6	A file uses contiguous allocation and occupies blocks 70 to 85. Answer the following questions- (a) How many blocks does the file occupy? (b) If the file grows by 5 blocks, can it still expand contiguously? Explain.	1.5+1.5	7	An	M

Section – B

5 X 3 = 15 Marks

Attempt All Questions

Attempt All Questions

No.	Detail of Question	Marks	CO	BL	KL																																		
7	Describe the hold and wait condition and how can it be prevented? An OS contains 3 resource classes. The number of resource units in these classes are 7,7,10 respectively. The Current resource allocation is shown below: <table><tr><th rowspan="2"></th><th colspan="3">Allocated Resource</th><th colspan="3">Maximum Requirement</th></tr><tr><th>R1</th><th>R2</th><th>R3</th><th>R1</th><th>R2</th><th>R3</th></tr><tr><td>P1</td><td>2</td><td>2</td><td>3</td><td>3</td><td>6</td><td>8</td></tr><tr><td>P2</td><td>2</td><td>0</td><td>3</td><td>4</td><td>3</td><td>3</td></tr><tr><td>P3</td><td>1</td><td>2</td><td>4</td><td>3</td><td>4</td><td>4</td></tr></table> Answer the following questions using Banker's Algorithm: a) Is the current allocation in safe state? b) Would the following independent requests be		Allocated Resource			Maximum Requirement			R1	R2	R3	R1	R2	R3	P1	2	2	3	3	6	8	P2	2	0	3	4	3	3	P3	1	2	4	3	4	4	3+1+1	5	A	P
	Allocated Resource			Maximum Requirement																																			
	R1	R2	R3	R1	R2	R3																																	
P1	2	2	3	3	6	8																																	
P2	2	0	3	4	3	3																																	
P3	1	2	4	3	4	4																																	

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	granted in current state: i) Process P1 request (1,1,0) ii) Process P2 request (0,1,0)				
8	(a) What are the necessary conditions for the deadlock? Explain with an example. (b) A computer system uses paging. The time taken to check the TLB is 10 nanoseconds, and the time to access main memory is 80 nanoseconds. The TLB hit ratio is 90% then calculate the effective memory access time.	3+2	6	R	F
9	A system has 3 frames. Page references are: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2 The replacement policy works as follows: • When a page fault occurs and a replacement is needed, the page that will not be used for the longest time in the future is removed. Identify the algorithm in this scenario and calculate the page fault.	1+4	6	An	P

Section – C

Attempt All Questions

4 X 3 = 12 Marks

No.	Detail of Question	Marks	CO	BL	KL
10	Describe the differences between symmetric and asymmetric multiprocessing. What are three advantages and one disadvantage of multiprocessor systems?	2+2	1	R	F
11	Consider the following set of processes, with the length of the CPU burst given in milliseconds: Process Burst Time Priority P1 2 2 P2 1 1 P3 8 4 P4 4 2 P5 5 3 The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0. a. Draw two Gantt charts that illustrate the	2+2	3	A	P

	execution of these processes using the following scheduling algorithms: SJF, non-preemptive priority (a larger priority number implies a higher priority). b. What is the turnaround time and waiting time of each process for each of the scheduling algorithms in part a?				
12	Consider a disk queue with requests for I/O to blocks on cylinders 30, 210, 455, 300, 499, 188, 367, 120, 280, 450, 470, 10. The cylinders are numbered from 0 to 499. The head is initially at cylinder number 250. The head moves only forward; after reaching the last cylinder, it jumps back to the lowest cylinder and continues servicing. 1. Identify the disk scheduling algorithm. 2. Determine servicing order and total seek movement.	1+3	7	An	M